

Shubham Kale

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EDUCATION

Johns Hopkins University

Expected May 2029

B.S. Applied Mathematics & Statistics, B.S. Computer Science, B.A. Economics | GPA: 3.70/4.00 | Baltimore, MD

- **Coursework:** Linear Algebra for Data Science, Statistical Analysis, Data Structures & Algorithms, Microeconomics.
- **Honors:** Hodson Trust Scholar (from 45,000+ applicants), 1For2 Foundation Scholar (5 of 1,000+)

EXPERIENCE

Summer Atlantic Capital LLC & SAC Nexus

Jun 2026 – Present

Machine Learning Engineer / Data Scientist, Product Development Team

Remote

- Built and shipped a compliance-scoring layer across **4 financial-risk workflows** (KYC, transactions, escrow, AML), cutting KYC case-reconstruction by **1–3 min/review** and transaction review by **1–2 min/payment** — roughly **4 analyst-hours/day** across 80 daily cases
- Developed an **AML candidate-ranking and explanation system** that compressed a 45-minute partner-screening workflow to **15–20 minutes (a 56–67% reduction)**, saving **4–5 analyst-hours per week per analyst**; modeled at **\$20K/year** of labor capacity per analyst, or **\$200K/year** across a 10-analyst deployment
- Built the modeling stack from an empty repository: a **5-pattern anomaly detector** (structuring, velocity, new-counterparty, amount-outlier, corridor) using **z-score deviation over a 7-day rolling baseline**, and a weighted **rules-based risk scorer** across 5 factors mapped to 3 severity bands; served behind a versioned **FastAPI inference service** with **114 tests** and CI at **80% branch coverage**
- Trained **node2vec graph embeddings** (gensim skip-gram, **64-dim**) with a versioned artifact registry and **64-d FAISS index**; engineered **40 ranking features** (16 topology, 19 dynamic, 5 embedding) feeding an **XGBoost LambdaMART ranker** (rank:ndcg) with per-prediction attribution

Open Health Intelligence (ClarityAtlas)

Sep 2025 – May 2026

Co-Founder & Technical Lead — Applied ML

Baltimore, MD

- Designed a **tappigraphy-based sleep-estimation model** inferring sleep windows from noisy phone-interaction and motion signals at **90% accuracy**; piloted with **300+ Johns Hopkins students**, earning a **non-binding \$75K investment indication** and **presentation to JHU President Ron Daniels and DSAI leadership**
- Built the **longitudinal data pipeline** end to end: PostgreSQL time-series schema, HealthKit/Core Motion ingestion under intermittent connectivity, a **weighted Euclidean similarity-matching service**, and **8 geofenced triggers**

Gates Foundation Health Equity Project (JHU PIER/PACER)

Sep 2025 – Jun 2026

Lead Model Developer — under Dr. Kate Rucinski

Baltimore, MD

- Identified primary drivers of multimorbidity across a **10M-record** national survey dataset using **XGBoost** with **SHAP-based variable selection**, grounded in the Heise et al. (2019) *Lancet* framework
- Designed **G-formula causal-inference simulations** estimating counterfactual outcomes under alternative interventions, informing policy recommendations for a Gates-funded JHU–SAMRC partnership

Intelehealth

Sep 2025 – Jun 2026

Software Engineering / AI Research Intern

Hybrid (Baltimore & India)

- Designed an **evaluation framework** (stratified sampling via Cochran’s formula) benchmarking LLM-generated outputs against a **4,169-case** gold standard, quantifying accuracy and surfacing **failure modes**; built a Python data pipeline across **32 sites**

PROJECTS

Dispatch — Workflow Automation Agent | TypeScript, Playwright, SQLite

2026

- Built a deterministic agent across **3 third-party platforms** with a **SQLite state machine** over **8 pipeline states**, **fail-closed flags** on every mutating capability, and a four-stage **validation ladder** requiring proof against live systems, not fixtures

tSearch — Talent Discovery Engine | TypeScript, Node, Neo4j, OpenAI

2026

- Shipped a working **MVP in 4 days** for a problem scoped by the founder of Z Fellows, iterating through **2 rounds of feedback** into an introduction to his team; built **2,000+ candidate profiles** into a navigable **Neo4j agent graph** with **rubric-driven LLM judges** producing evidence-cited scores

TECHNICAL SKILLS

Languages: Python (pandas, NumPy, scikit-learn, XGBoost, SHAP, PyTorch), R, SQL, TypeScript/JavaScript, C/C++, Java

Modeling: CNNs, RNN/Transformer sequence models, gradient boosting (XGBoost, LambdaMART), node2vec graph embeddings, learning-to-rank, anomaly detection, causal inference (G-formula), RAG, vector search

Methods & Infra: experimental design, stratified sampling, feature selection, model evaluation, failure-mode analysis; FastAPI, PostgreSQL, Neo4j, Pinecone, FAISS, AWS, Git/CI